

Accident Data Research Summary

Accident Data Sources Successfully Collected

1. United States (Nationwide)

- **Source:** Fatality Analysis Reporting System (FARS) by the National Highway Traffic Safety Administration (NHTSA)
- **Access:** API available
- **Coverage:** All U.S. states
- **Data Quality:** Includes complete accident details along with accurate latitude and longitude coordinates
- **Use Case:** Selected and integrated into the system for high-quality and continuous access

2. United Kingdom (Nationwide)

- **Source:** UK Government Road Safety Data
- **Link:** <https://www.data.gov.uk/dataset/cb7ae6f0-4be6-4935-9277-47e5ce24a11f/road-accidents-safety-data>
- **Access:** API available (notably accessible via R)
- **Coverage:** All regions of the United Kingdom (data from 2018 onwards)
- **Data Quality:** Well-structured, with spatial coordinates and comprehensive fields

3. South Africa (Nationwide)

- **Source:** South Australia Government Open Data Portal – Road Crash Data
- **Link:** <https://data.sa.gov.au/data/dataset/road-crash-data>
- **Access:** API available
- **Coverage:** Nationwide data available from 2018 onwards
- **Data Quality:** Full geographic coverage with coordinate-level granularity

Data Sources Explored but Not Utilized

The following sources were thoroughly evaluated. However, due to missing geographic data, lack of API access, limited geographic coverage, or formatting inconsistencies, they were not selected for use in the current phase.

1. Japan

- **Source:** Institute for Traffic Accident Research and Data Analysis (ITARDA)
- **Link:** <https://www.itarda.or.jp/english/map>

- **Issue:** The dataset does not include geographic coordinates and is not accessible via API

2. New York City, USA

- **Source:** Vision Zero NYC
- **Link:** <https://vzv.nyc/>
- **Issue:** No support for API-based access. Although historical data is available (up to 2024), the records do not consistently include latitude/longitude values

3. Canada (Nationwide)

- **Sources:**
 - <https://www.toronto.ca/services-payments/streets-parking-transportation/road-safety/vision-zero/safety-measures-and-mapping/>
 - <https://open.canada.ca/data/en/dataset/1eb9eba7-71d1-4b30-9fb1-30cbdab7e63a>
 - **Issue:** National-level data sources either do not include geographic coordinates or are not available in a consistent, machine-readable format. No suitable API was identified for full-country coverage
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City-Level Canadian Datasets (Reviewed for Potential Future Use)

While complete nationwide data for Canada could not be acquired, several city-level datasets were discovered. These sources may be considered for future expansions or region-specific modules.

- **Ottawa**
 - Link: <https://open.ottawa.ca/datasets/ottawa%3A%3Atraffic-collision-data/about>
- **Halifax**
 - Link: <https://data-hrm.hub.arcgis.com/datasets/traffic-collisions/explore>
- **Calgary**
 - Link: https://data.calgary.ca/Transportation-Transit/Traffic-Incidents/35ra-9556/about_data

These datasets are limited to individual municipalities but offer high-resolution local data, which may complement the core system in future localized deployments.

Conclusion

The research phase successfully identified robust and API-supported datasets for the United States, United Kingdom, and South Africa. Each selected source meets the minimum requirements for:

- Complete geographic coverage
- Availability of spatial coordinates
- API-based data access
- Reliable historical records

Other sources, while reviewed, were excluded based on data gaps or technical limitations. This ensures the Alert Prediction System is built upon high-quality, scalable, and dependable data assets. Further integration of localized datasets can be considered based on project needs or specific client regions.